Lista 6 - MNED

1. Mostre que o erro de truncamento do método

$$U_j^{n+1} = U_j^n + \sigma(U_{j-1}^n - 2U_j^n + U_{j+1}^n)$$

que aproxima o problema exato $u_t = \kappa u_{xx}$, é dado por

$$\tau_j^n = \frac{\Delta t}{2} u_{tt}(x_j, t_n) - \frac{\kappa \Delta x^2}{12} u_{xxxx}(x_j, t_n) + O(\Delta t^2) + O(\Delta x^4).$$

Em seguida, encontre o valor de σ (o que significa impor condições sobre Δt e Δx) que faça com que este método tenha ordem de consistência $O(\Delta t^2)$.

$$\tau_j^n = \frac{u(x_j, t_n+\Delta t) - u(x_j, t_n)}{\Delta t} - \frac{\sigma}{\Delta t}\left(\Delta x^2 u_{xx} + \frac{\Delta x^4}{12} u_{xxxx} + O(\Delta x^6)\right)$$

$$u_{xxxx} = \frac{\partial^2}{\partial x^2}\left(\frac{\partial^2 u}{\partial x^2}\right) = \frac{1}{k}\frac{\partial^2}{\partial x^2} u_t = \frac{1}{k^2} u_{tt}$$

$$= \frac{\partial u}{\partial t} + \frac{\Delta t}{2}\frac{\partial^2 u}{\partial t^2} + \frac{\Delta t^2}{6}\frac{\partial^3 u}{\partial t^3} + O(\Delta t^3) - \frac{\sigma \Delta x^2}{\Delta t\, k} u_t - \frac{\sigma \Delta x^4}{12 \Delta t\, k^2} u_{tt} + \frac{\sigma}{\Delta t} O(\Delta x^6)$$

se $\sigma = \dfrac{\Delta t\, k}{\Delta x^2}$

$$= \frac{\partial u}{\partial t} - \frac{\partial u}{\partial t} + \frac{\partial^2 u}{\partial t^2}\left(\frac{\Delta t}{2} - \frac{\Delta x^2}{k 12}\right) + O(\Delta t^2) + O(\Delta x^4)$$

$$= \frac{\partial^2 u}{\partial t^2}\left(\frac{\Delta t}{2} - \frac{\Delta x^2}{12}\right) + O(\Delta t^2) + O(\Delta x^4)$$

Para termos o método $O(\Delta t^2)$, devemos ter:

$$\frac{\Delta t}{2} - \frac{\Delta x^2}{k 12} = 0 \Rightarrow \Delta x^2 = 6 \Delta t\, k$$

$$\Rightarrow \sigma = \frac{\Delta t\, k}{\Delta x^2} = \frac{\Delta t\, k}{6 \Delta t\, k} = \frac{1}{6}$$

2. Mostre que o método explícito

$$U_j^{n+1} = \frac{2\sigma}{3} U_{j+1}^n + (1-2\sigma) U_j^n + \frac{4\sigma}{3} U_{j-1}^n$$

não é consistente com a equação $u_t = \kappa u_{xx}$. Encontre a equação para a qual esse método é consistente.

$$U_j^{n+1} = U_j^n + 2\sigma\left(\frac{1}{3}U_{j+1}^n - U_j^n + \frac{2}{3}U_{j-1}^n\right) \qquad \sigma = \frac{\Delta t\, k}{\Delta x^2}$$

$$\tau_j^n = u_t + \frac{\Delta t}{2} u_{tt} + O(\Delta t^2) - \frac{2\sigma}{\Delta t}\left[\frac{1}{3}\left(u + \Delta x u_x + \frac{\Delta x^2}{2} u_{xx} + \frac{\Delta x^3}{6} u_{xxx} + \frac{\Delta x^4}{24} u_{xxxx}\right) - u + O(\Delta x^5) + \frac{2}{3}\left(u - \Delta x u_x + \frac{\Delta x^2}{2} u_{xx} - \frac{\Delta x^3}{6} u_{xxx} + \frac{\Delta x^4}{24} u_{xxxx}\right)\right]$$

$$= u_t + \frac{\Delta t}{2} u_{tt} + O(\Delta t^2) - \frac{2\sigma}{\Delta t}\left[-\frac{\Delta x}{3} u_x + \frac{\Delta x^2}{2} u_{xx} - \frac{\Delta x^3}{18} u_{xxx} + \frac{\Delta x^4}{24} u_{xxxx} + O(\Delta x^5)\right]$$

$$= u_t + \frac{\Delta t}{2} u_{tt} + O(\Delta t^2) + \frac{2 \Delta t\, k}{\Delta t \Delta x^2}\cdot\frac{\Delta x}{3} u_x - \frac{2\Delta t\, k}{\Delta t \Delta x^2}\cdot\frac{\Delta x^2}{2}\cdot\frac{1}{k} u_t + \frac{2\Delta t\, k}{\Delta t \Delta x^2}\cdot\frac{\Delta x^3}{18}\frac{1}{k} u_{xt} +$$

$$- \frac{2\Delta t\, k}{\Delta t \Delta x^2}\,\frac{\Delta x^4}{24}\,\frac{1}{k^2} u_{tt} + \frac{2\Delta t\, k}{\Delta t \Delta x^2}\cdot O(\Delta x^5)$$

$$= u_t(1-1) + u_{tt}\left(\frac{\Delta t}{2} - \frac{\Delta x^2}{12 k}\right) + \frac{2k}{3\Delta x} u_x + \frac{\Delta x}{9} u_{xt} + O(\Delta x^3) + O(\Delta t^2)$$

$$= u_{tt}\left(\frac{\Delta t}{2} - \frac{\Delta x^2}{12 k}\right) + \underbrace{\frac{2k}{3\Delta x}}_{\Downarrow} u_x + \frac{\Delta x}{9} u_{xt} + O(\Delta x^3) + O(\Delta t^2)$$

o método não é consistente com $u_t = k u_{xx}$

$$u_t + \frac{\Delta t}{2} u_{tt} + O(\Delta t^2) - \frac{2\sigma}{\Delta t}\left[-\frac{\Delta x}{3} u_x + \frac{\Delta x^2}{2} u_{xx} - \frac{\Delta x^3}{18} u_{xxx} + \frac{\Delta x^4}{24} u_{xxxx} + O(\Delta x^5)\right]$$

Analisando novamente o erro de truncamento:

$$\tau_j^n = u_t + \frac{\Delta t}{2} u_{tt} + O(\Delta t^2) - \frac{2\Delta t\, k}{\Delta t \Delta x^2}\left[-\frac{\Delta x u_x}{3} + \frac{\Delta x^2}{2} u_{xx} - \frac{\Delta x^3}{18} u_{xxx} + \frac{\Delta x^4}{24} u_{xxxx} + O(\Delta x^5)\right] =$$

$$= u_t + \frac{\Delta t}{2} u_{tt} + \frac{2}{3\Delta x} k u_x - k u_{xx} + \frac{\Delta x k u_{xxx}}{9} - k\frac{\Delta x^2}{12} u_{xxxx} + O(\Delta t^2) + O(\Delta x^3)$$

Dessa forma se $u_t - k u_{xx} + \frac{2}{3\Delta x} u_x = 0$, o método é consistente $\Rightarrow$ esse método resolve a EDP

$$u_t = k\left(u_{xx} - \frac{2}{3\Delta x} u_x\right)$$

3. Discuta a consistência da generalização do método de DuFort-Frankel ($0 \le \theta \le 1$)

$$\frac{U_j^{n+1} - U_j^{n-1}}{2\Delta t} = \kappa \frac{U_{j+1}^n - 2(\theta U_j^{n+1} + (1-\theta)U_j^{n-1}) + U_{j-1}^n}{\Delta x^2},$$

para os casos: (a) $\Delta t = r\kappa \Delta x$, e (b) $\Delta t = r\kappa \Delta x^2$, onde r é uma constante positiva.

Centrando a expansão de Taylor em (x_j, t_n):

$$\tau_j^n = \frac{1}{2\Delta t}\left(\cancel{u(x_j,t_n)} + \Delta t\, u_t(x_j,t_n) + \cancel{\frac{\Delta t^2}{2} u_{tt}(x_j,t_n)} + \frac{\Delta t^3}{6} u_{ttt}(x_j,t_n) + \cancel{\frac{\Delta t^4}{24} u_{tttt}} - \cancel{u(x_j,t_n)} + \Delta t\, u_t(x_j,t_n) - \cancel{\frac{\Delta t^2}{2} u_{tt}(x_j,t_n)} + \frac{\Delta t^3}{6} u_{ttt}(x_j,t_n) - \cancel{\frac{\Delta t^4}{24} u_{tttt}}\right) + O(\Delta t^4)$$

$$-\frac{k}{\Delta x^2}\Big[u(x_j,t_n) + \cancel{\Delta x\, u_x(x_j,t_n)} + \frac{\Delta x^2}{2} u_{xx}(x_j,t_n) + \cancel{\frac{\Delta x^3}{6} u_{xxx}(x_j,t_n)} + \frac{\Delta x^4}{24} u_{xxxx}(x_j,t_n) +$$

$$u(x_j,t_n) - \cancel{\Delta x\, u_x(x_j,t_n)} + \frac{\Delta x^2}{2} u_{xx}(x_j,t_n) - \cancel{\frac{\Delta x^3}{6} u_{xxx}(x_j,t_n)} + \frac{\Delta x^4}{24} u_{xxxx}(x_j,t_n) -$$

$$2\theta\left(u(x_j,t_n) + \Delta t\, u_t(x_j,t_n) + \frac{\Delta t^2}{2} u_{tt}(x_j,t_n) + \frac{\Delta t^3}{6} u_{ttt}(x_j,t_n) + \frac{\Delta t^4}{24} u_{tttt}(x_j,t_n)\right)$$

$$-2(1-\theta)\left(u(x_j,t_n) - \Delta t\, u_t + \frac{\Delta t^2}{2} u_{tt} - \frac{\Delta t^3}{6} u_{ttt} + \frac{\Delta t^4}{24} u_{tttt}\right) + O(\Delta x^6)\ O(\Delta t^5)\Big]$$

$$= u_t + \frac{\Delta t^2}{6} u_{ttt} + O(\Delta t^4)$$

$$-\frac{k}{\Delta x^2}\Big[u(2-2-2\theta+2\theta) + \Delta x^2 u_{xx} + \frac{\Delta x^4}{12} u_{xxxx}$$

$$+ \Delta t\, u_t(-2\theta + 2 - 2\theta) + \frac{\Delta t^2}{2} u_{tt}(-2\theta - 2 + 2\theta) +$$

$$+ \frac{\Delta t^3}{6} u_{ttt}(-2\theta + 2 - 2\theta) + \frac{\Delta t^4}{24}(-2\theta - 2 + 2\theta)\Big]$$

$$= u_t + \frac{\Delta t^2}{6} u_{ttt} + O(t^4) - k\Big[u_{xx} + \frac{\Delta x^2}{12} u_{xxxx} +$$

$$+ \frac{\Delta t}{\Delta x^2} u_t(-4\theta+2) + \frac{\Delta t^2}{2\Delta x^2} u_{tt}(-2) + \frac{\Delta t^3}{6} u_{ttt}(-4\theta+2) + \frac{\Delta t^4}{24}(-2)\Big] + O(\Delta x^4)$$

$$= \underbrace{u_t - k u_{xx}}_{} + \frac{\Delta t^2}{6} u_{ttt} - k\frac{\Delta x^2}{12} u_{xxxx} + O(\Delta x^4) + O(\Delta t^4)$$

$$+ k(4\theta-2)\frac{\Delta t}{\Delta x^2} u_t + k\frac{\Delta t^2}{\Delta x^2} u_{tt} + \frac{(2\theta-1)k}{3}\frac{\Delta t^3}{\Delta x^2} u_{ttt} + \frac{k \Delta t^4}{12} u_{tttt}$$

0 na eq do calor

a) $\Delta t = r k \Delta x$

$$\tau_j^n = \frac{\Delta t^2}{6} u_{ttt} - k\frac{\Delta x^2}{12} u_{xxxx} + 2k(2\theta-1)\frac{\Delta t}{\Delta x^2} u_t + k\frac{\Delta t^2}{\Delta x^2} u_{tt} + \frac{(2\theta-1)k}{3}\frac{\Delta t^3}{\Delta x^2} u_{ttt} + k\frac{\Delta t^4}{12} u_{tttt} + O(\Delta x^4) + O(\Delta t^4)$$

$$= \frac{k^2r^2\Delta x^2}{6}u_{ttt} - k\frac{\Delta x^2}{12}u_{xxxx} + \frac{2k(2\theta-1)rk\cancel{\Delta x}}{\Delta x^{\cancel{2}}} u_t + \frac{r^2k^3\cancel{\Delta x^2}}{\cancel{\Delta x^2}}u_{tt} + \frac{(2\theta-1)k^4r^3\Delta x^{\cancel{3}}}{\cancel{\Delta x^2}}u_{ttt} + \frac{k^5r^4\Delta x^4}{12}u_{tttt} + O(\Delta x^4)$$

se $2\theta-1=0 \Rightarrow \theta=\frac{1}{2}$, o termo $\frac{2k^2(2\theta-1)r}{\Delta x}u_t = 0$. Logo, o método não é consistente, independente de θ, pois, no melhor caso, é $O(r^2k^3u_{tt}) = O(cte)$

b) $\Delta t = rk\Delta x^2$

$$\tau_j^n = \frac{r^2k^2\Delta x^4}{6}u_{ttt} - \frac{kr\Delta x^2}{12}u_{xxxx} + \frac{2(2\theta-1)k^2r\cancel{\Delta x^2}}{\cancel{\Delta x^2}}u_t + \frac{k^2r^2\Delta x^{\cancel{4}\,2}}{\cancel{\Delta x^2}}u_{tt} + \frac{(2\theta-1)k^4r^3\Delta x^{\cancel{6}\,4}}{3\cancel{\Delta x^2}}u_{ttt} + \frac{k^5r^4\Delta x^8}{12}u_{tttt} + O(x^4)$$

Se $\theta=\frac{1}{2}$, o método é $O(\Delta x^2)$, caso contrário, não é consistente

4. Considere a equação diferencial

$$u_t = u_{xx} - bu, \quad t>0, \quad x\in\mathbb{R}$$

com $b>0$. Analise a convergência do método de diferenças finitas:

$$U_j^{n+1} = \sigma U_{j-1}^n + (1-2\sigma - b\Delta t)U_j^n + \sigma U_{j+1}^n.$$

$\sigma = \frac{\Delta t}{\Delta x^2}$

Reescrevendo:

$$U_j^{n+1} - U_j^n = \sigma(U_{j-1}^n + U_{j+1}^n) - (2\sigma + b\Delta t)U_j^n$$

$$\Rightarrow \frac{U_j^{n+1}-U_j^n}{\Delta t} = \frac{1}{\Delta x^2}(U_{j-1}^n - 2U_j^n + U_{j-1}^n) - bU_j^n$$

Consistência:

$$\tau_j^n = \frac{u(x_j,t_n+\Delta t)-u(x_j,t_n)}{\Delta t} - \frac{1}{\Delta x^2}\left(u(x_j-\Delta x,t_n) - 2u(x_j,t_n) + u(x_j+\Delta x,t_n)\right) + bu(x_j,t_n) =$$

Expandindo em Taylor centrado em (x_j,t_n):

$$= u_t + \frac{\Delta t}{2}u_{tt} + \frac{\Delta t^2}{6}u_{ttt} + O(\Delta t^3) - \left(u_{xx} + \frac{\Delta x^2}{12}u_{xxxx} + O(\Delta x^4)\right) + bu$$

$$= \underbrace{u_t - u_{xx} + bu}_{0} + \frac{\Delta t}{2}u_{tt} + \frac{\Delta t^2}{6}u_{ttt} - \frac{\Delta x^2}{12}u_{xxxx} + O(\Delta t^3) + O(\Delta x^4)$$

$\Rightarrow \tau_j^n = O(\Delta t + \Delta x^2)$

O método é consistente.

Estabilidade:

Como estamos em um problema de Cauchy, podemos usar análise de Von-Neumann sem preocupações. Nesse caso,

$$\frac{U_j^{n+1}-U_j^n}{\Delta t} = \frac{1}{h}(U_{j-1}^n - 2U_j^n + U_{j-1}^n) - bU_j^n$$

fazemos $U_j^n = e^{ijh\xi}$, logo $U_j^{n+1} = g(\xi)e^{ijh\xi}$. Encontremos restrições tq. $|g(\xi)| \le 1 + \alpha\Delta t$. Substituindo no método:

$$g(\xi)e^{ijh\xi} = e^{ijh\xi} + \frac{\Delta t}{h^2}\left(e^{i(j-1)h\xi} - 2e^{ijh\xi} + e^{i(j+1)h\xi}\right) - \Delta t b e^{ijh\xi}$$

$$\Rightarrow g(\xi) = 1 + \frac{\Delta t}{h^2}(e^{-ih\xi} + e^{ih\xi} - 2) - \Delta t b = 1 + \frac{2\cos(h\xi)-2}{h^2}\Delta t - \Delta t b$$

Como $-1\le\cos(h\xi)\le 1$, temos

$$1 - \frac{4\Delta t}{h^2} - \Delta t b \le g(\xi) \le 1 - \Delta t b \qquad b>0$$

Precisamos apenas que $-1 \le 1 - 4\frac{\Delta t}{h^2} - \Delta t b$

$\Rightarrow -2 \le -4\sigma - \Delta t b \Rightarrow 4\sigma + \Delta t b \le 2$

$\therefore$ O método tem estabilidade forte quando $4\sigma + \Delta t b \le 2$

$\Rightarrow$ Converge para $4\sigma + \Delta t b \le 2$

5. Considere a equação diferencial

$$\begin{aligned} u_t &= au_{xx} - bu_x, \quad a>0,\quad 0\le x\le L,\quad 0\le t\le T,\\ u(x,0) &= \psi(x), \quad 0\le x\le L,\\ u(0,t)&=f(t),\quad 0\le t\le T,\\ u(L,t) &= g(t), \quad 0\le t\le T.\end{aligned}$$

(a) Mostre que discretizando essa equação pelo método explícito e diferenças centrais para aproximar u_x, obtém-se

$$U_j^{n+1} = \left(\sigma + \frac{b\Delta t}{2\Delta x}\right)U_{j-1}^n + (1-2\sigma)U_j^n + \left(\sigma - \frac{b\Delta t}{2\Delta x}\right)U_{j+1}^n$$

(b) Calcule o erro de truncamento local;
(c) Utilizando o critério de von Neumann, estude a estabilidade desse método;
(d) Modifique o método acima, aproximando o termo u_x pelo chamado método *upwind*, que consiste em discretizar u_x por diferenças progressivas se $b<0$ e por diferenças regressivas se $b>0$;
(e) Calcule o erro de truncamento local do novo método, e analise a sua estabilidade.

$$u_t(x_j,t_n) = \frac{u(x_j,t_n+\Delta t)-u(x_j,t_n)}{\Delta t} + O(\Delta t) \approx \frac{U_j^{n+1}-U_j^n}{\Delta t}$$

$u_{xx}(x_j,t_n) \approx \frac{U_{j-1}^n - 2U_j^n + U_{j+1}^n}{\Delta x^2}$ $\quad (O(\Delta x^2))$

$u_x(x_j,t_n) \approx \frac{U_{j+1}^n - U_{j-1}^n}{2\Delta x}$ $\quad (O(\Delta x^2))$

Temos o método: $\qquad \sigma = \frac{a\Delta t}{\Delta x^2}$

$$\frac{U_j^{n+1}-U_j^n}{\Delta t} = \frac{a}{\Delta x^2}(U_{j+1}^n - 2U_j^n + U_{j-1}^n) - \frac{b}{2\Delta x}(U_{j+1}^n - U_{j-1}^n)$$

$$\Rightarrow U_j^{n+1} = U_j^n + \sigma(U_{j+1}^n - 2U_j^n + U_{j-1}^n) - \frac{b\Delta t}{2\Delta x}(U_{j+1}^n - U_{j-1}^n)$$

$$\Rightarrow U_j^{n+1} = \left(\sigma - \frac{b\Delta t}{2\Delta x}\right)U_{j+1}^n + (1-2\sigma)U_j^n + \left(\sigma+\frac{b\Delta t}{2\Delta x}\right)U_{j-1}^n$$

b) O único erro vem das discretizações $\Rightarrow$ $\tau_j^n = O(\Delta t + \Delta x^2)$. Mó rolê calcular as constantes do erro!

c) Encontremos o módulo do fator de amplificação $g(\xi)$. Como no ex 4:

$$g(\xi)e^{ijh\xi} = \left(\sigma - \frac{b\Delta t}{2\Delta x}\right)e^{i(j+1)h\xi} + (1-2\sigma)e^{ijh\xi} + \left(\sigma + \frac{b\Delta t}{2\Delta x}\right)e^{i(j-1)h\xi}$$

$$\Rightarrow g(\xi) = \left(\sigma - \frac{b\Delta t}{2\Delta x}\right)e^{ih\xi} + (1-2\sigma) + \left(\sigma+\frac{b\Delta t}{2\Delta x}\right)e^{-ih\xi}$$

$$= \sigma(e^{ih\xi}+e^{-ih\xi}) - \frac{b\Delta t}{2\Delta x}(e^{ih\xi}-e^{-ih\xi}) + (1-2\sigma) = 2\sigma\cos(h\xi) - \frac{b\Delta t}{2\Delta x}2i\operatorname{sen}(h\xi)$$

$$+ 1-2\sigma$$

$\mu = \frac{b\Delta t}{\Delta x} \Rightarrow g(\xi) = 2\sigma\cos(h\xi) - \mu i \,\mathrm{sen}(h\xi) + 1 - 2\sigma$

$= 1 + 2\sigma(\cos(h\xi) - 1) - i\mu\,\mathrm{sen}(h\xi)$

$|g(\xi)|^2 = (1 + 2\sigma(\cos(h\xi)-1))^2 + \mu^2\,\mathrm{sen}^2(h\xi)$

$1 - \cos(h\xi) = 1 - \left(\cos^2\left(\frac{h\xi}{2}\right) - \mathrm{sen}^2\left(\frac{h\xi}{2}\right)\right) =$

$= 1 - \cos^2\left(\frac{h\xi}{2}\right) + \mathrm{sen}^2\left(\frac{h\xi}{2}\right)$

$= 2\,\mathrm{sen}^2\left(\frac{h\xi}{2}\right)$

$\mathrm{sen}^2(h\xi) = \left(2\,\mathrm{sen}\left(\frac{h\xi}{2}\right)\cos\left(\frac{h\xi}{2}\right)\right)^2 = 4\,\mathrm{sen}^2\left(\frac{h\xi}{2}\right)\left(1 - \mathrm{sen}^2\left(\frac{h\xi}{2}\right)\right)$

$s = \mathrm{sen}\left(\frac{h\xi}{2}\right) \Rightarrow \mathrm{sen}^2(h\xi) = 4s^2(1-s^2)$

$1 - \cos(h\xi) = 2s^2$ (tirado do GPT)

$\Rightarrow |g(\xi)|^2 = (1 - 4\sigma s^2)^2 + 4\mu^2 s^2(1-s^2)$

$= 1 + 16\sigma^2 s^4 - 8\sigma s^2 - 4\mu^2 s^2(s-1)(s+1) \leq 1$

$\Rightarrow 16\sigma^2 s^4 - 8\sigma s^2 - 4\mu^2 s^2(s^2-1) \leq 0$

$\Rightarrow 16\sigma^2 s^2 - 8\sigma - 4\mu^2(s^2-1) \leq 0$

$\Rightarrow 4\sigma^2 s^2 - 2\sigma - \mu^2 s^2 + \mu^2 \leq 0 \Rightarrow \underbrace{(\mu^2 - 2\sigma) + s^2(4\sigma^2 - \mu^2)}_{} \leq 0$

crescente/decrescente pois $s^2 \in [0,1]$. Precisamos apenas que os extremos sejam ≤ 0.

$s^2 = 0 \Rightarrow \mu^2 - 2\sigma \leq 0 \Rightarrow \mu^2 \leq 2\sigma$

$s^2 = 1 \Rightarrow 4\sigma^2 - \mu^2 \leq 2\sigma - \mu^2 \Rightarrow 4\sigma^2 - 2\sigma \leq 0 \Rightarrow \sigma \leq \frac{1}{2}$

$\therefore \mu^2 \leq 2\sigma$ e $\sigma \leq \frac{1}{2}$

d) $u_x \approx \frac{U_{j+1} - U_j}{\Delta x}$ ou $u_x \approx \frac{U_j - U_{j-1}}{\Delta x}$

Por simplicidade, faremos apenas para $b<0 \Rightarrow$ diferenças progressivas

$$U_j^{n+1} = \left(\sigma + \frac{b\Delta t}{\Delta x}\right) U_{j+1}^n + \left(1 - 2\sigma - \frac{b\Delta t}{\Delta x}\right) U_j^n + \sigma U_{j-1}^n$$

e) $O(\Delta t + \Delta x)$

Estabilidade:

$g(\xi) = (\sigma + \mu)e^{ih\xi} + (1 - 2\sigma - \mu) + \sigma e^{-ih\xi}$

$|g(\xi)| = |(\sigma+\mu)e^{ih\xi} + (1 - 2\sigma - \mu) + \sigma e^{-ih\xi}| \leq$

$\leq |\sigma + \mu|\,\underbrace{|e^{ih\xi}|}_{=1} + |1 - 2\sigma - \mu| + |\sigma|\,\underbrace{|e^{-ih\xi}|}_{=1} =$

$= |\sigma + \mu| + |1 - 2\sigma - \mu| + |\sigma|$

se $\sigma, \mu \geq 0$ $\quad |g(\xi)| \leq \sigma + \mu + 1 - 2\sigma - \mu + \sigma = 1$

6. Considere o seguinte problema de valor de fronteira de Neumann

$$\begin{aligned} u_t &= \kappa u_{xx}, \quad \kappa > 0, \quad 0 \leq x \leq L, \quad 0 \leq t \leq T, \\ u(x,0) &= \psi(x), \quad 0 \leq x \leq L, \\ u_x(0,t) &= \alpha_1(u-1), \quad 0 \leq t \leq T, \\ u_x(L,t) &= -\alpha_2(u-1), \quad 0 \leq t \leq T \end{aligned}$$

onde α_1 e α_2 são constantes positivas.
(a) Obtenha o método resultante da aproximação das condições de fronteira por diferenças centrais e da equação pelo método explícito;
(b) Obtenha o método resultante da aproximação da fronteira esquerda por diferenças progressivas da fronteira direita por diferenças regressivas e da equação pelo método explícito;
(c) Repetir os itens acima considerando o método implícito para aproximar a equação.

Escreva esse método em notação matricial. Qual o efeito das condições de fronteira na matriz dos coeficientes? Usando o segundo teorema de Gershgorin estude a estabilidade do método resultante do primeiro item, mostrando que a restrição de estabilidade é dada por:

$$\sigma \leq \min\left\{\frac{1}{2+\alpha_1\Delta x}, \frac{1}{2+\alpha_2\Delta x}\right\}.$$

a) $U_j^{n+1} = U_j^n + \sigma(U_{j-1}^n - 2U_j^n + U_{j+1}^n)$ $\qquad \sigma = \frac{\kappa\Delta t}{\Delta x^2}$

em $j = 0 \Rightarrow x_0 = 0$

$u_x(x_0, t_n) = \alpha_1(u(x_0,t_n) - 1) \Rightarrow \frac{U_1^n - U_{-1}^n}{2\Delta x} = \alpha_1(U_0^n - 1)$ $\qquad O(\Delta x^2)$

$\Rightarrow -U_{-1}^n = -U_1^n + 2\alpha_1\Delta x(U_0^n - 1) \Rightarrow U_{-1}^n = U_1^n - 2\alpha_1\Delta x(U_0^n - 1)$

$\Rightarrow U_0^{n+1} = U_0^n + \sigma(U_1^n - 2\alpha_1\Delta x(U_0^n - 1) - 2U_0^n + U_1^n) \Rightarrow$

$$\boxed{U_0^{n+1} = (1 - 2\alpha_1\Delta x\sigma - 2\sigma)U_0^n + 2\sigma U_1^n + 2\alpha_1\Delta x\sigma}$$

em $j = m \Rightarrow x_{m+1} = L$

$u_x(x_m, t_n) = -\alpha_2(u - 1) \Rightarrow \frac{U_{m+1}^n - U_{m-1}^n}{2\Delta x} = -\alpha_2(U_m^n - 1)$ $\qquad O(\Delta x^2)$

$\Rightarrow U_{m+1}^n = U_{m-1}^n - 2\alpha_2\Delta x(U_m^n - 1)$

$\Rightarrow U_m^{n+1} = U_m^n + \sigma(U_{m-1}^n - 2U_m^n + U_{m-1}^n - 2\alpha_2\Delta x(U_m^n - 1)) \Rightarrow$

$$\boxed{U_m^{n+1} = (1 - 2\alpha_2\Delta x - 2\sigma)U_m^n + 2\sigma U_{m-1}^n + 2\alpha_2\Delta x\sigma}$$

Notação matricial:

$\underline{U}^{n+1} = A\underline{U}^n + \underline{g}^n$

$$A = \begin{bmatrix} 1 - 2\sigma(\alpha_1\Delta x + 1) & 2\sigma & & & 0 \\ \sigma & (1-2\sigma) & \sigma & & \\ & \ddots & \ddots & \ddots & \\ & & \sigma & (1-2\sigma) & \sigma \\ 0 & & & 2\sigma & 1 - 2\sigma(\alpha_2\Delta x + 1) \end{bmatrix} \qquad g^n = \begin{bmatrix} 2\alpha_1\Delta x\sigma \\ 0 \\ \vdots \\ 0 \\ 2\alpha_2\Delta x\sigma \end{bmatrix}$$

Precisamos que $\|A\|_2 = \rho(A) \leq 1$

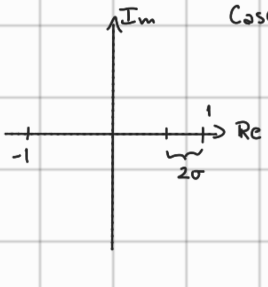
Discos de Gershgorin:

$R_i = 2\sigma, \quad i = 0, \dots, m$

$\Rightarrow D(a_{ii}, R_i) = D(1 - 2\sigma, 2\sigma), \quad i = 1, \dots, m-1$

$D(a_{00}, R_0) = D(1 - 2\sigma(\alpha_1\Delta x + 1), 2\sigma)$

$D(a_{mm}, R_m) = D(1 - 2\sigma(\alpha_2\Delta x + 1), 2\sigma)$



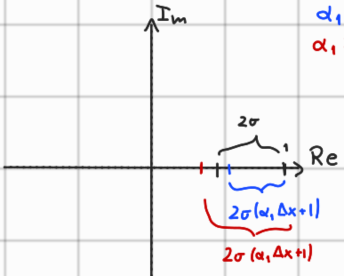
Caso $i = 1, \dots, m-1$

$2\sigma \leq 1$

$\sigma \leq \frac{1}{2} \Rightarrow \frac{\kappa\Delta t}{\Delta x^2} \leq \frac{1}{2}$

$i=0$

$\sigma > 0$



$\alpha_1 < 0$

$\alpha_1 > 0$

$\Rightarrow$ Não há garantia de estabilidade

$|1-2\sigma(\alpha_1\Delta x+1)+2\sigma| \leq 1$

$|1-2\sigma(\alpha_1\Delta x+1)-2\sigma| \leq 1$

$\Rightarrow -1 \leq 1-2\sigma\alpha_1\Delta x \leq 1 \Rightarrow -2 \leq -2\sigma\alpha_1\Delta x \leq 0 \Rightarrow 2\sigma\alpha_1\Delta x \leq 2 \Rightarrow \sigma \leq \frac{1}{\alpha_1\Delta x}$

$-1 \leq 1-2\sigma(\alpha_1\Delta x+2) \leq 1 \Rightarrow -1 \leq -\sigma(\alpha_1\Delta x+2) \leq 0 \Rightarrow \sigma \leq \frac{1}{2+\alpha_1\Delta x} \leq \frac{1}{\alpha_1\Delta x}$

$\therefore \sigma \leq \frac{1}{2+\alpha_1\Delta x}$

Da mesma forma, $\sigma \leq \frac{1}{2+\alpha_2\Delta x}$

$\Rightarrow \sigma \leq \min\left\{\frac{1}{2+\alpha_1\Delta x}, \frac{1}{2+\alpha_2\Delta x}\right\}$

7. Seja $\delta_x^2 U_j^n = U_{j+1}^n - 2U_j^n + U_{j-1}^n$, e considere o método obtido pela média ponderada dos métodos explícito e implícito

$$U_j^{n+1} = U_j^n + \sigma\left(\theta\delta_x^2 U_j^{n+1} + (1-\theta)\delta_x^2 U_j^n\right)$$

com $\sigma = \kappa\Delta t/\Delta x^2$ e $0 \leq \theta \leq 1$. Esta expressão define os métodos da família θ. Note que $\theta = 0$ resulta no método explícito, $\theta = 1$ resulta no método implícito, e $\theta = \frac{1}{2}$ resulta no método de Crank-Nicolson.
(a) Deduza o erro de truncamento local para este método;
(b) Utilizando a técnica de von Neumann, mostre que para $0 \leq \theta < \frac{1}{2}$, o método é condicionalmente estável se $\sigma \leq \frac{1}{2(1-2\theta)}$ e incondicionalmente estável para $\frac{1}{2} \leq \theta \leq 1$.